

ETHAN PHAM

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EDUCATION

University of California, San Diego

Expected Graduation: June 2028

Bachelor of Science in Computer Science

GPA: 4.0/4.0

Relevant Coursework: Accelerated Introduction to Java Programming, Data Structures and Object-Oriented Design, Discrete Mathematics, Systems Programming and Software Tools in C

SKILLS

Languages, Tools, & Frameworks: Python, Java, JavaScript, C, VSCode, vim, Git/GitHub

EXPERIENCE

National Honors Society

Lake Forest, CA

STEM Tutor

Dec 2022 - May 2024

- Provided targeted academic support to 5+ students across AP Computer Science, Physics, and Calculus by breaking down complex concepts into clear, step-by-step explanations to improve understanding and retention.
- Increased students' homework completion rates by 30% by identifying individual learning gaps and guiding them through problem-solving strategies during lunchtime sessions.
- Fostered a collaborative learning environment by proactively engaging students who were hesitant to seek help, leading to higher participation and confidence in STEM subjects.

AI Bootcamp

Remote

Developer

Jun 2023 - Jul 2023

- Developed a web-based article summarization tool by leveraging JavaScript, HTML, CSS, Flask, and Python, enabling users to generate concise summaries from input text, improving reading efficiency by 50%.
- Spearheaded weekly brainstorming sessions and daily stand-ups to facilitate transparent team communication, resulting in faster consensus on design decisions and smoother collaboration.
- Implemented frontend components and integrated backend APIs to deliver a fully functional application within a 3-week sprint, demonstrating strong time management and cross-functional teamwork.

El Toro High School

Lake Forest, CA

Community Service Volunteer

Jan 2023

- Collaborated with a team of 10+ volunteers to decorate and serve at a community pancake breakfast for over 150 attendees, ensuring smooth operations and maintaining a welcoming atmosphere for families.
- Facilitated efficient cleanup and waste management post-event by coordinating trash collection and venue restoration, reducing teardown time by 15%.
- Communicated effectively with event organizers and peers to align on tasks and priorities, contributing to high attendee satisfaction and a successful community gathering.

PROJECTS

Pioneer Shell

May 2025

- Developed a custom shell emulation supporting built-in commands (exec, cd, history) by implementing input parsing and command dispatch logic in C.
- Utilized fork() and execvp() system calls to execute external programs dynamically, enabling the shell to run non-hardcoded processes.
- Built interactive and batch processing modes to accept user commands from either keyboard input or batch files, increasing flexibility for different workflows.

Malloc Simulator

Apr 2025

- Engineered a 16-bit memory alignment system supporting dynamic allocation and deallocation through block splitting and coalescing, recreating C's malloc/free functionality.
- Designed header and footer metadata structures within memory blocks to track allocation status and optimize traversal and coalescing performance.

Unix Calendar

Mar 2025

- Created a Unix time-based one-week calendar using a linked list backend to store and manage scheduled events.
- Implemented user-facing features for event creation, deletion, searching, rescheduling, duplication, and memory cleanup, improving usability and data integrity.